Viraj Karambelkar

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Education

PhD, California Institute of Technology 2021-2025
Astrophysics

MS, California Institute of Technology 2019-2021
Astrophysics

B.Tech, Indian Institute of Technology, Bombay 2015-2019
Engineering Physics with Honors, Minor in Mathematics

Professional Experience

NASA Hubble Fellow Sep. 2025 - present
Columbia University, USA

Graduate Research Assistant 2019-2025
Caltech, USA

Caltech Visiting Undergraduate Research Program (VURP) Fellow Summer 2018
Caltech, USA

Undergraduate Research Assistant 2017-2019
IIT Bombay, India

Awards

Hubble Fellowship NASA (2025)

Neugebauer Scholar of Astrophysics, Caltech, USA (2023-2025)

Finalist, 3-Minute Thesis Competition, Caltech (2024)

Institute Academic Prize for exceptional academic performance at IIT Bombay. (2017)

Successful Proposals (Telescope and Grants)

NASA Hubble Fellowship

- The Anthropology of Stellar Mergers (Awarded $321,993)

James Webb Space Telescope

- PI : 18 hours, Cycles 5, 6, and 7 : *An Elusive Dust Factory at Work : Tracing real-time dust and molecule formation in a giant star merger*, (Awarded $100,000)
- PI : 4.9 hours, Cycle 4 DDT : *First Shift at an Elusive Dust Factory: Early dust and molecule formation in a giant star merger*, (Awarded $35,000)
- PI : 3.2 hours, Cycle 4 DDT : *CEE it happen: Common-Envelope Evolution in action with early JWST observations of a stellar merger in M31*, (Awarded $35,000)
- PI : 10.4 hours, Cycle 2 : *Are LRNe Major factories of cosmic dust?*, (Awarded $80,000)

Hubble Space Telescope

- PI : 2 orbits Cycle 30 : *In search of the remnant of SN 2021fcg – detonation, deflagration or merger?*, (Awarded $10,000)
- co-I : SNAP program Cycle 29 : *UV Spectroscopy of Astronomical Transients through Rolling Snapshots*

LSST Discovery Alliance 2026

- Marrying the deep with the red : An open source tool for crossmatching NEOWISE alerts with Rubin (Awarded $12,000)

Gemini

- PI : 7 hours, Director's Discretionary Time 2025B : *GMOS followup of a stellar merger in M31*

NASA - Infrared Telescope Facility

- PI : 2 nights, 2026A : *Uncovering hidden stellar mergers in the Milky Way with NEOWISE, SPHEREx and IRTF*
- PI : 2 nights, 2025B : *Characterizing bright infrared outbursts with NEOWISE, SPHEREx and IRTF*
- PI : 2 nights, 2022B : *Telling them apart - Identifying the first chemical differences between R Coronae Borealis and dustless HdC stars*
- PI : 4 nights 2021B : *An Infrared census of R Coronae Borealis stars*

Palomar 200-inch telescope

- PI: 2 nights 2023B : *Completing the census of large amplitude variable stars identified by Palomar-Gattini IR*
- co-I: 35 nights (2021-2023) : *The Dynamic Infrared Sky*

Keck I+II telescopes

- co-I: 20 nights (2022-2024) : *Census of the local universe with ZTF*

Gran Telescopio Canarias (GTC)

- co-I: (2023) *Multi-wavelength target of opportunity follow-up of newly-discovered luminous red novae*

Very Large Array (VLA)

- co-I: 6 hours DDT : *Chasing a very bright GRB at VLA frequencies - GRB 230812B*
- co-I: 1 hour DDT : *IRAS 19148+1138: an Asymptotic Giant Branch star candidate in VLASS*

Sub-mm Array (SMA)

- Awarded 4 hours as part of the SMA-Interferometry School 2021 for the proposal *Tracing molecular gas in the envelope of R Coronae Borealis*

Swift telescope

- Total 30 ks of approved ToO time for early time UV followup of transients.

PUBLICATIONS

Full list here. Total refereed: 85, Total citations: 3000; h-index:30

Select publications with major contributions

- **V. Karambelkar**, K. De, L. Hillenbrand et al. "SPHEREx as a Frontier for Infrared Transients: Classification of New Galactic FU Ori Outbursts and Classical Novae" In ApJ Letters (February 2026)
- **V. Karambelkar**, M. Kasliwal, R. Lau et al. "Hot Springs and Dust Reservoirs: JWST Reveals the Dusty, Molecular Aftermath of Extragalactic Stellar Mergers" In ApJ (January 2026)
- **V. Karambelkar**, M. Kasliwal, K. De et al. "The Slow Brightening of WNTR23bzdiq/WTP19aalzlk: Possible Onset of Common Envelope Evolution in an Asymptotic Giant Branch Star?" In ApJ (November 2025)
- **V. Karambelkar**, M. Kasliwal, P. Tisserand et al. "Census of R Coronae Borealis Stars II : Spectroscopic classifications and implications for the rate of low mass white-dwarf mergers" In PASP (July 2024)
- **V. Karambelkar**, M. Kasliwal, N. Blagorodnova et al. "Volumetric Rates of Luminous Red Novae and Intermediate-luminosity Red Transients with the Zwicky Transient Facility" In ApJ (May 2023)
- A. Suresh [1], **V. Karambelkar**, M. Kasliwal et al. "An automated catalog of long-period variables from Palomar Gattini IR" In PASP (April 2024)
- A. Mehla [1], M. Kasliwal, **V. Karambelkar**, et al. "Oxygen Isotope Ratios in Hydrogen-Deficient Carbon Stars: A Correlation with Effective Temperature and Implications for White Dwarf Merger Outcomes" In PASP (April 2024)
- K. De, M. MacLeod, **V. Karambelkar** et al. "An infrared transient from a star engulfing a planet". In Nature 617.7959 (May 2023)
- H. Tranin, N. Blagorodnova, **V. Karambelkar** et al. "Hertzsprung gap stars in nearby galaxies and the Quest for Luminous Red Novae Progenitors" submitted to A&A (Sep. 2024)
- G. Dimitriadis, K. Maguire, **V. Karambelkar** et al. "SN 2021zny: an early flux excess combined with late-time oxygen emission suggests a double white dwarf merger event" In MNRAS (May 2023)
- **V. Karambelkar**, M. M. Kasliwal, P. Tisserand et al. "R Coronae Borealis and dustless hydrogen-deficient carbon stars likely have different oxygen isotope ratios". In A& A (Nov. 2022)
- D. Frostig, S. Biscoveanu, G. Mo, **V. Karambelkar** et al. "An Infrared Search for Kilonovae with the WINTER Telescope. I. Binary Neutron Star Mergers". In ApJ (Feb. 2022)
- **V. Karambelkar**, M. Kasliwal, K. Maguire et al. "Faintest of Them All: ZTF 21aaoryiz/SN 2021fcg-Discovery of an Extremely Low Luminosity Type Iax Supernova". In ApJ (Nov. 2021)
- M. Dhuria, **V. Karambelkar**, V. Rentala, and P. Sarmah "A strong broadband 21 cm cosmological signal from dark matter spin-flip interactions". In JCAP (Aug 2021)

[1] mentored

- **V. Karambelkar**, M. Kasliwal, P. Tisserand et al. "Census of R Coronae Borealis Stars. I. Infrared Light Curves from Palomar Gattini IR". In ApJ (Apr. 2021)
- N. Blagorodnova, **V. Karambelkar**, S. M. Adams et al. "Progenitor, precursor, and evolution of the dusty remnant of the stellar merger M31-LRN-2015". In MNRAS (Aug. 2020)
- **V. Karambelkar**, S. M. Adams, P. A. Whitelock et al. "SPIRITS Catalog of Infrared Variables: Identification of Extremely Luminous Long Period Variables". In ApJ (June 2019)

Software skills

Advanced proficiency in Python, C++, SQL, HTML, git, Tensorflow and PyTorch.

Select open source softwares :

- mirar : Image processing software used by WINTER, SEDMv2 and DREAMS surveys.
- kowalski : An alert broker for time-domain astronomy.
- pydusty : MCMC-software for modeling dust around transients.

Service, Mentorship and Outreach

Peer review

- Reviewer for ApJ, AJ, A&A, MNRAS, Nature Astronomy
- Time Allocation Committee, NOIRLab (2025 - 2026)
- Proposal reviewer for 2025 Fondecyt Research Initiation Competition, Ministry of Science, Chile

Teaching

- TA: Graduate courses Interstellar Medium, Radiative Processes, Radio Astronomy (Caltech, 2021)
- TA: Undergrad courses on Electromagnetism and Calculus (IIT Bombay, 2016, 2019)
- Lecturer: ZTF Summer School (2021)
- Tutor: GROWTH Summer Schools (2018, 2020)

Mentorship

- Graduate Students : Ayanna Mann (Columbia), Nicholas Earley (Caltech)
- Columbia Undergrads : Maria Cuevas
- Caltech SURF 2024 : Advait Mehla, Physics Postbacc, IIT Bombay
  Project : *Measuring elemental abundances for RCB and dlHdC stars* (paper published)
- Caltech SURF 2023 : Aswin Suresh, Engineering Physics Junior, IIT Bombay
  Project: *An ML-based catalog of Long Period Variables from Palomar Gattini IR* (paper published)
- Caltech SURF 2022 : Sulekha Kishore, Computer Science Sophomore, Caltech
  Project: *Developing an alert-broker for WINTER* (paper with contributions published)
- Caltech SURF 2021 : Kayton Truong, Computer Science Freshman, Caltech
  Project: *Searching for periodic variables in Palomar Gattini IR* (paper with contributions published)
- Department Academic Mentor, IIT Bombay (2017 - 2019)

Outreach

- **Astrobites writer** (2020-2022) Published 13 articles summarizing research papers at a level accessible to undergraduates.
- Co-manager - Cahill Rooftop Observatory, Caltech (2021 - 2023)
- Volunteer at public lectures and stargazing events organized by Caltech Astro Outreach.
- Head, Students Association Physics Department, IIT Bombay (2017-2018)

TALKS, CONFERENCES AND WORKSHOPS

Invited talks

- Rochester Institute of Technology Colloquium, USA (2026)
- Carnegie Observatories Seminar, Pasadena, USA (2024)
- Caltech/IPAC Seminar, Pasadena, USA (2024)
- MIT Astrophysics Seminar, Boston, USA (2024)
- Columbia University THEA Seminar, NYC, USA (2024)
- Center for Computational Astrophysics Seminar, NYC, USA (2024)
- NASA Goddard Space Flight Center Seminar, Greenbelt, USA (2024)
- Space Telescope Science Institute Seminar, Baltimore, USA (2024)
- University of Barcelona Seminar, Barcelona, Spain (2024)
- Princeton University Seminar, Princeton, USA (2023)
- Raman Research Institute Colloquium, Bangalore, India (2023)
- International Center for Theoretical Studies, Bangalore, India (2023)
- Indian Institute of Astrophysics, Bangalore, India (2023)
- Tata Institute of Fundamental Research (2023)
- Inter-University Center for Astronomy and Astrophysics (2021)

Contributed Talks and Posters

- Talk - 360° approach to common envelope evolution - University of Barcelona (2024)
- Talk - Symposium on common envelope evolution, EAS meeting, Krakow (2023)
- Talk - The Transient and Variable Universe, UIUC (2023)
- Talk - White dwarfs from physics to astrophysics, KITP, UCSB (2022)
- Talk - 240th meeting of the American Astronomical Society, Pasadena (2023)
- Talk - Zwicky Transient Facility Team Meeting, Paris (2022)
- Talk - Exploring the transient universe with the Roman Space Telescope, Pasadena (2022)
- Poster - Keck Science Meeting 2021, San Diego (2021)
- Poster - Super Virtual Conference (2021)
- Talk - 235th meeting of the American Astronomical Society, Hawaii (2020)
- Poster - GROWTH conference, Mumbai (2018)
- Poster - Nobel Prize Lectures, IIT Bombay, Mumbai (2017)

Workshops

- Sub-mm Array Interferometry School for sub-mm astronomy (2021)
- ComSciCon Workshop for science communication, Los Angeles (2021)
- ZTF Summer Undergraduate Astronomy School, Pasadena (2018)
- National Initiative for Undergraduate Studies Physics Camp, Mumbai (2016)
- National Talent Search Scholars Nurturance Camp, Mumbai (2012)